CSC317 Computer Graphics

Term Test 1 Review

Fall 2026

Solutions, grading rubric, and common mistakes

Topics

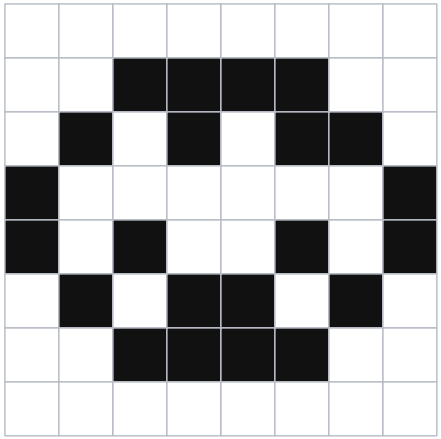
Q1 Raster images
Difference functions and 8 × 8 matching

Q2 Ray–cone intersection
Tangency, surface normals, and intersection counts

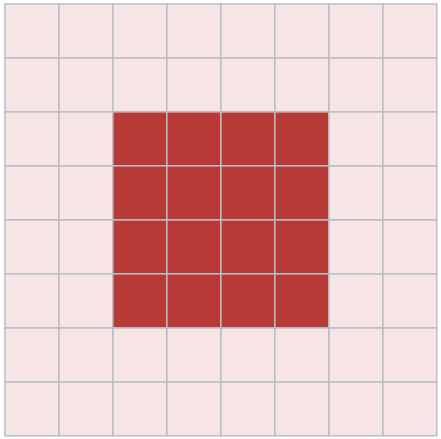
Q3 Ray tracing
Specular reflection, refraction, and secondary rays

Q1 Comparing two small images

E: gray level

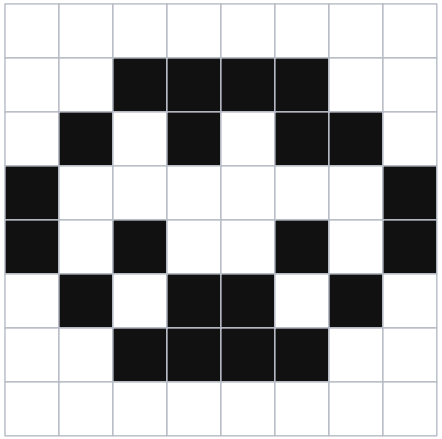


E: importance α



pale: α = 0 red: α = 1

I: gray level



E is an 8 × 8 emoticon. Its gray levels and α lie in [0, 1].

I has gray levels only. α belongs to E.

The stated setup has at least one pixel with α = 1.

Q1(a) Difference function

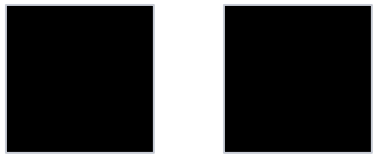
2 marks

Define f(E, I) for every pair of 8 × 8 images.

The value must lie in [0, 1].

$f(E, I) = 0$	when E and I are identical
$f(E, I) = 1$	when all pixels are maximally different, weighted by α

Q1(a) Pixel differences and importance



$|0 - 0| = 0$



$|0 - 1| = 1$



$|0.3 - 0.7| = 0.4$

Weight each difference by its importance:

$\alpha_p \, |E_p - I_p|$

Q1(a) A weighted mean difference

$$f(E, I) = \frac{\sum_p \alpha_p \, |E_p - I_p|}{\sum_p \alpha_p}$$

Identical images: every difference is 0, so f = 0.

Maximal differences: each weighted difference equals α_p, so the numerator equals the denominator and f = 1.

If every $\alpha_p = 0$, define f = 0 to avoid division by zero.
The rubric did not deduct marks for omitting this edge case.

Q1(a) Grading

2 marks

2	Returns 0 for identical images and 1 for maximally different images, with α weighting.
Accepted	Other image pairs may also return 0 or 1. The required endpoints still have to be correct.
−1	Otherwise correct, but does not use α.
−0.5	Otherwise correct, but assumes an α for I.
−0.5	Missing or incorrect normalization.
0	Defines the function only in the two extreme cases.
No penalty	Weighted averages should address all $\alpha = 0$.

Q1(b) Difference of averages

2 marks

A simpler function compares only image averages:

$$g(E, I) = \left|\text{mean}(E) - \text{mean}(I)\right|$$

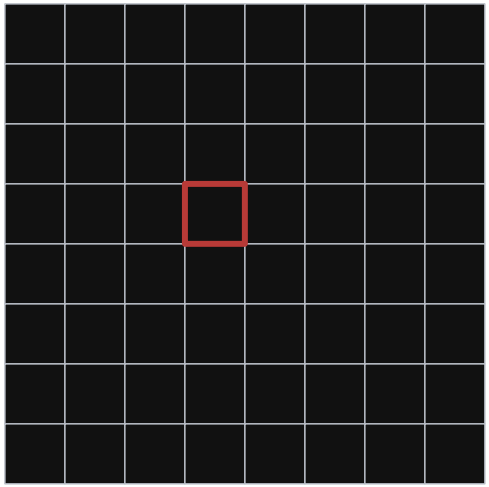
Construct images for each case:

(i) $f = 0$, $g = 63/64$

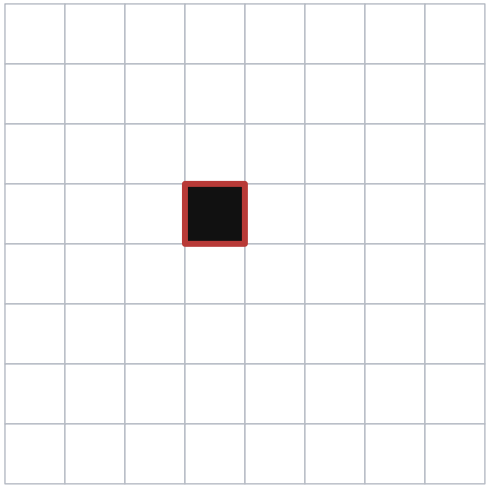
(ii) $f = 1$, $g = 0$

Q1(b)(i) f = 0, g = 63/64

E: every pixel is black



I: only pixel p is black



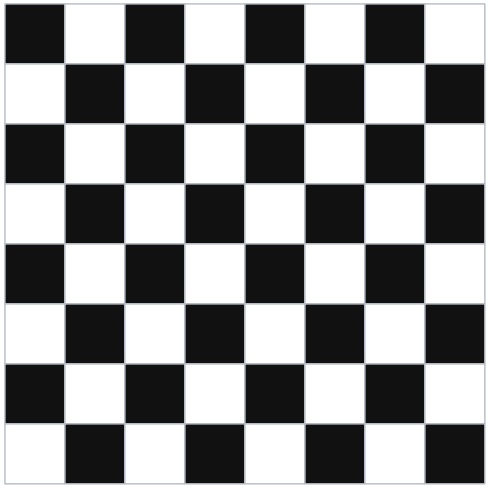
$\alpha_p = 1$ at the red-outlined pixel. All other $\alpha = 0$.

f = 0 because the only weighted pixel matches.

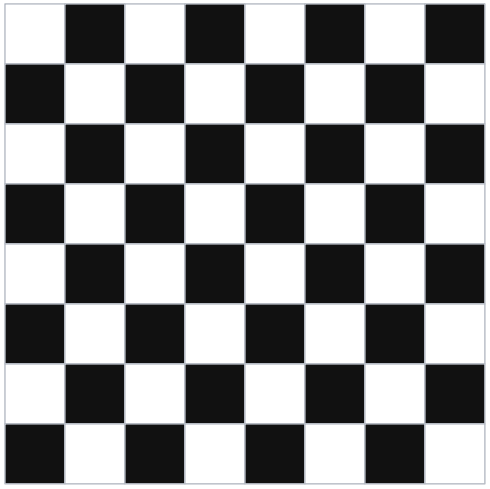
mean(E) = 0 and mean(I) = 63/64, so g = 63/64.

Q1(b)(ii) f = 1, g = 0

E: checkerboard



I: inverted checkerboard



Set α = 1 at every pixel of E.

Every pixel differs by 1, so f = 1.

Both images have average 1/2, so g = 0.

Q1(b) Grading

2 marks

1 each	A clear construction for each requested pair of f and g values.
Clarity	The description must not allow an incorrect E or I.
Follow-on	If Q1(a) was incorrect, Q1(b) was graded using the function you gave in Q1(a).

Specify E, I, and the α values used by f.

Q1(c) Finding every matching region

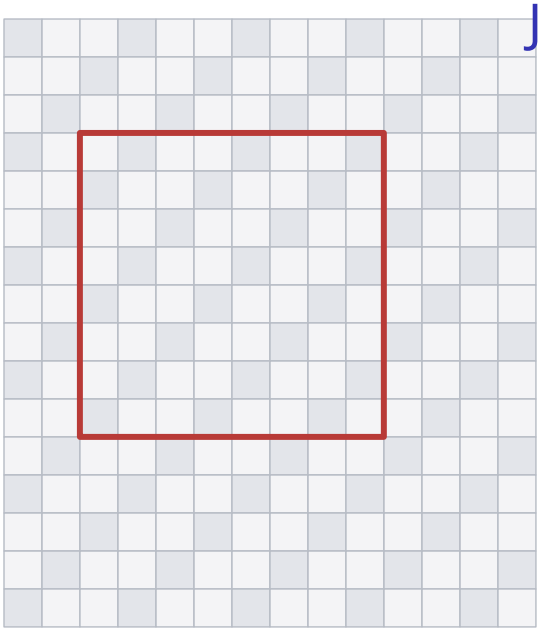
2 marks

J has m rows and n columns.

Test each 8 × 8 subimage W.

A match satisfies f(E, W) < e.

Return all matches or their count.



8 × 8 window W

Q1(c) Valid window positions

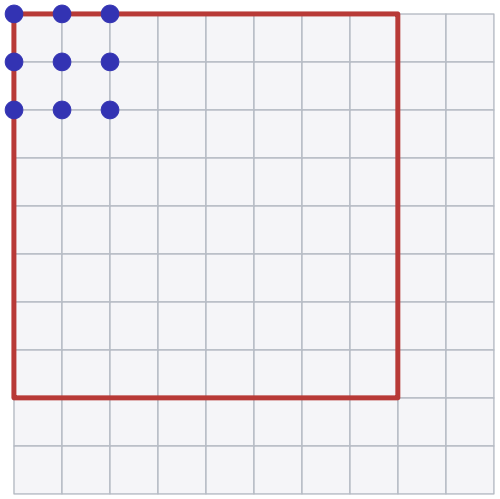
For zero-based row i and column j:

$$0 \le i \le m - 8, \quad 0 \le j \le n - 8$$

10 × 10 image: 3 × 3 = 9 windows.

8 × 8 image: exactly one window.

If either dimension is less than 8, there are no windows.



9 possible top-left corners

Q1(c) Sliding-window algorithm

```
matches = []
for i in range(m - 7):
    for j in range(n - 7):
        W = J[i:i+8, j:j+8]
        if f(E, W) < e:
            matches.append((i, j))
return matches
```

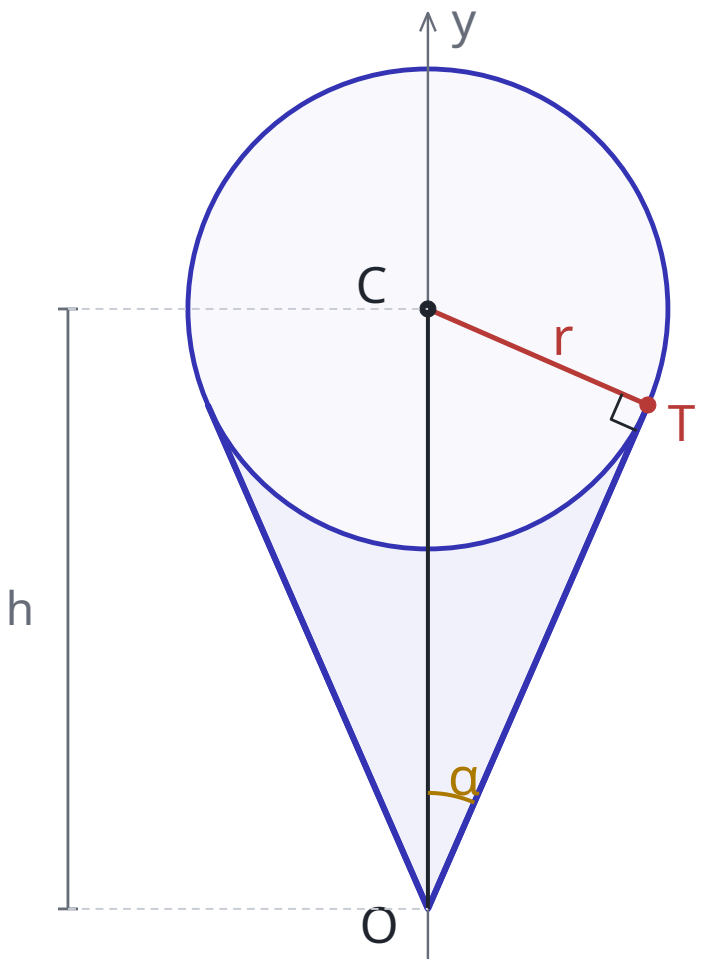
Python-style ranges exclude the upper endpoint.
For m = 8, range(m − 7) contains only i = 0.

Q1(c) Grading

2 marks

1	Explicit loop or recursion over 8 × 8 windows. No explicit sliding-window logic: no marks.
0.5 / 1	Sliding-window logic has boundary errors.
1	Checks f(E, window) < e.
0.5 / 1	Energy check occurs but another check nullifies it, e.g. testing energy only for identical windows.
0.5 / 1	Uses < with a specified threshold other than e.
−0.5	Returns only one match, or does not return.
Accepted	Returns a count instead of matching windows.

Q2 Cone tangent to a sphere



Cone apex: O = (0, 0, 0).

Sphere centre: C = (0, h, 0).

Sphere radius: r, with h > r.

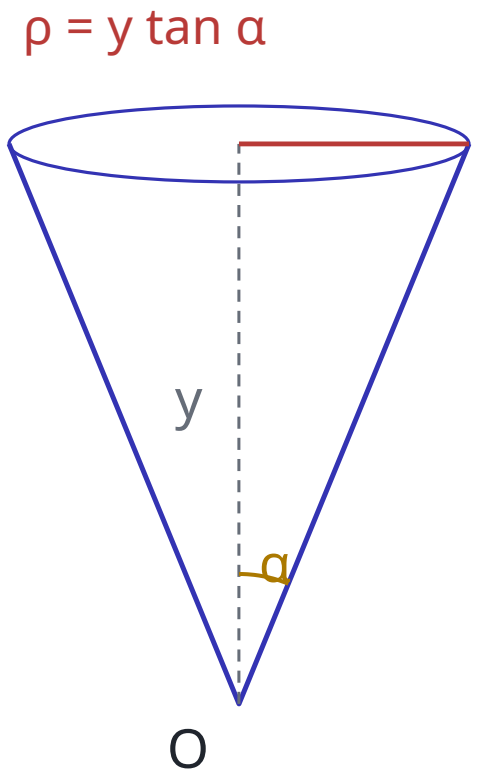
T is the tangency point.
CT is perpendicular to OT.

Q2 Cone equation

$$F(x, y, z) = x^2 + z^2 - y^2 \tan^2\alpha = 0$$

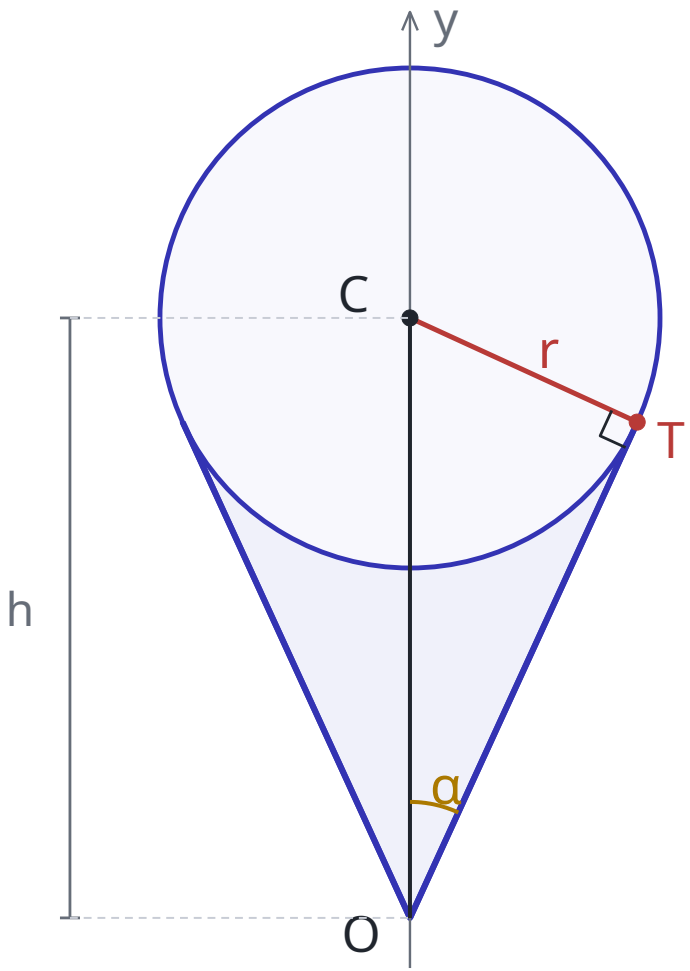
At height y, the radius is $\rho = y \tan \alpha$.

The circular cross-section satisfies $x^2 + z^2 = \rho^2$.



Q2(a) Cone half-angle

1 mark



OCT is right-angled at T.

Its hypotenuse is OC = h.
The side opposite α is CT = r.

$$\sin \alpha = r / h$$

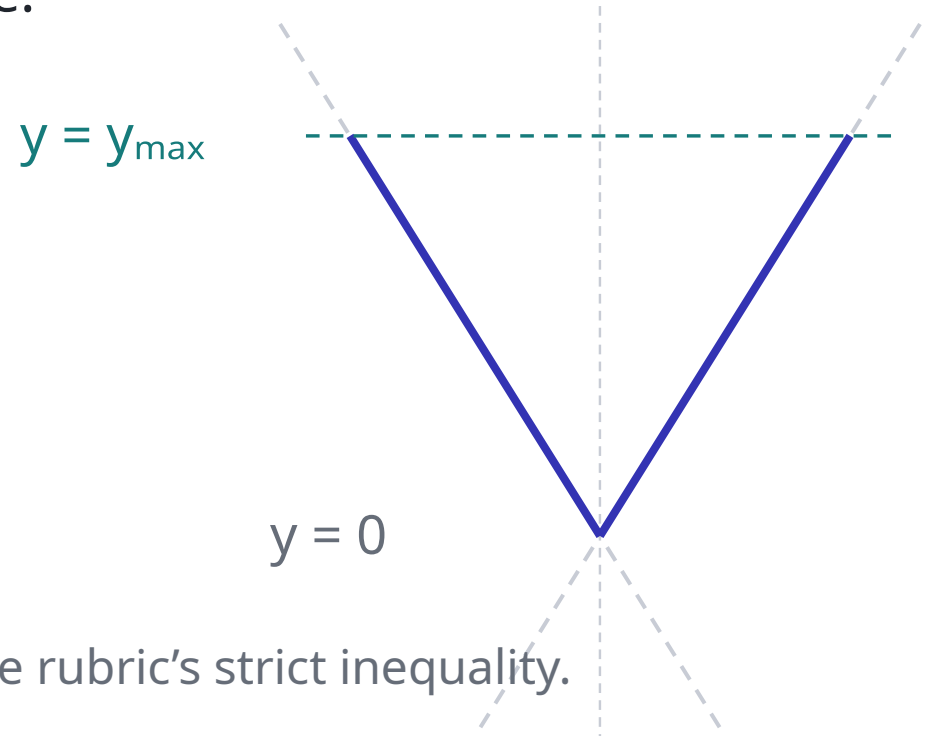
$$\alpha = \arcsin(r / h)$$

Q2(b) Finite cone surface

1.5 marks

F = 0 describes an infinite double cone.

The required surface starts at the apex and stops at the tangency circle.

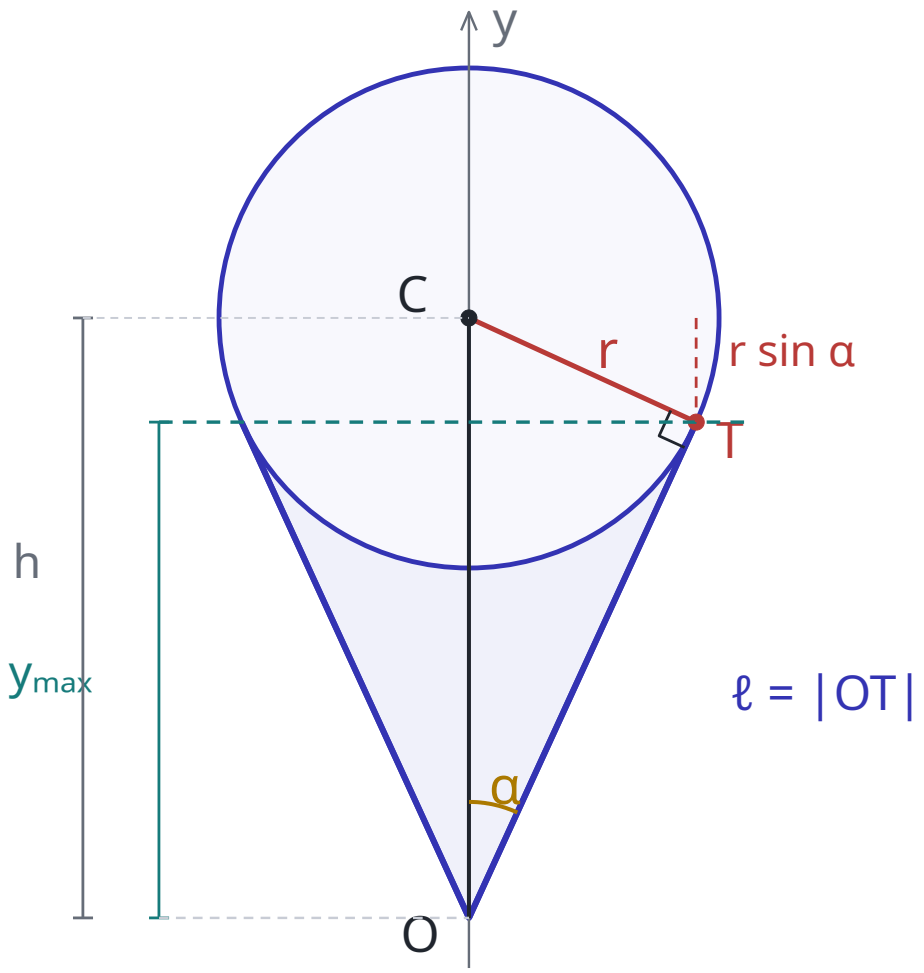


finite upper cone

$0 \leq P.y < y_{max}$

The upper boundary follows the rubric’s strict inequality.

Q2(b) Height at tangency



Slant length to T:

$$\ell = \sqrt{(h^2 - r^2)}$$

$$\cos\alpha = \ell / h$$

Vertical projection of OT:

$$y_{max} = \ell \cos\alpha$$

$$= h \cos^2\alpha$$

$$= h - r^2/h$$

$$= h - r \sin\alpha$$

Q2(a)–(b) Grading

1 Q2(a): $\alpha = \arcsin(r/h)$.
$\sin\alpha = r/h$ also accepted.

1.5 Q2(b): the tangency height, in an equivalent form:

$$P.y < h - r\sin\alpha$$

$$P.y < h\cos^2\alpha$$

$$\|P\| < \sqrt{(h^2 - r^2)}$$

The norm form applies to points on the upper cone.

0.5 Q2(b): $P.y < h$, or an equivalent partial check.

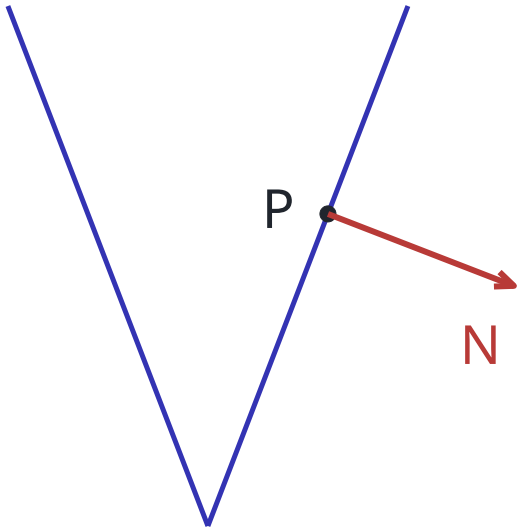
Q2(c) Cone surface normal

1 mark

$$F = x^2 + z^2 - y^2 \tan^2\alpha$$

$$\nabla F = (2x, -2y \tan^2\alpha, 2z)$$

$$N = (P.x, -P.y \tan^2\alpha, P.z)$$



∇F is perpendicular to the surface.
An unnormalized normal is sufficient.

At the apex, the smooth surface normal is undefined.

Q2(c) Grading

1 mark

1	Derives the gradient of the implicit cone equation.
Accepted	$N = \langle P.x, -P.y \tan^2\alpha, P.z \rangle$
	$N = P - \langle 0, P.y/\cos^2\alpha, 0 \rangle$
	$N = P - \langle 0, \|P\|/\cos\alpha, 0 \rangle$
0.5	Arithmetic error, incomplete derivation, or only stating “take the gradient”.
Accepted	Unnormalized N.

Q2(d) Cone intersection

4 marks

```
coneIntersect(S, D, α, r, h, t) -> int
```

Ray: $P(t_r) = S + t_r D$.

Find every forward hit on the finite cone surface.

Append valid parameters to the caller’s list t.

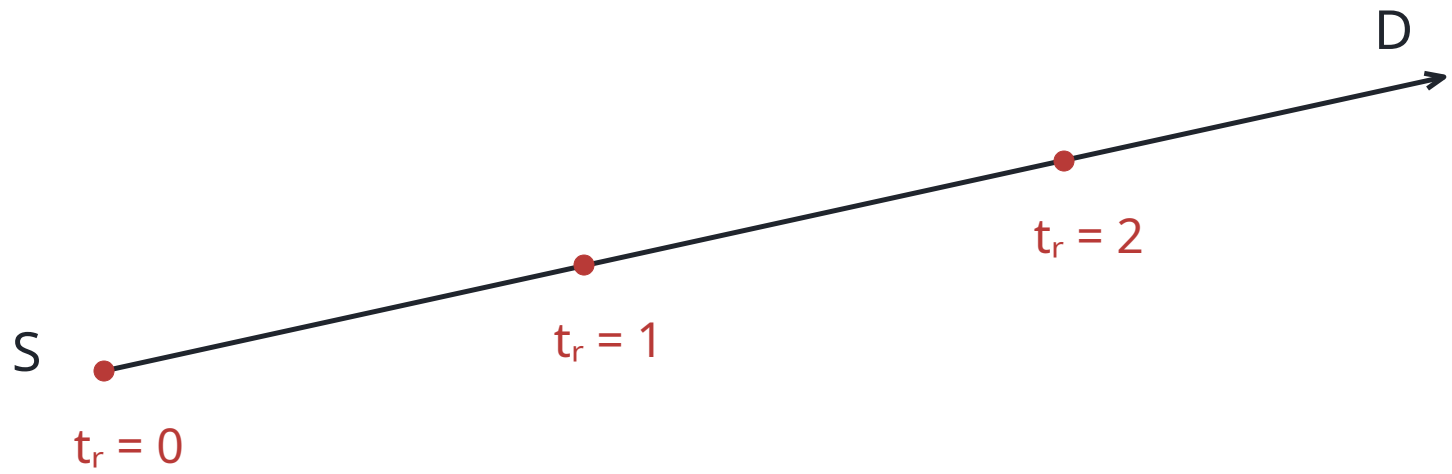
Sort t from nearest to furthest and return the count.

The list is an output parameter. Keep any existing hits when using it to combine sphere and cone intersections.

Q2(d) Substituting the ray

A surface intersection satisfies $F(S + t_r D) = 0$.

$$(S.x + t_r D.x)^2 + (S.z + t_r D.z)^2 - \tan^2\alpha\,(S.y + t_r D.y)^2 = 0$$



Q2(d) Quadratic coefficients

Let $k = \tan^2\alpha$.

$$A = D.x^2 + D.z^2 - k\, D.y^2$$

$$B = 2(S.x\, D.x + S.z\, D.z - k\, S.y\, D.y)$$

$$C = S.x^2 + S.z^2 - k\, S.y^2$$

$$A\, t_r^2 + B\, t_r + C = 0$$

$$\Delta = B^2 - 4AC$$

For $A \neq 0$: $\Delta < 0$ gives no real roots.
Otherwise $t_r = (-B \pm \sqrt{\Delta})/(2A)$.

Q2(d) Filtering the candidate roots

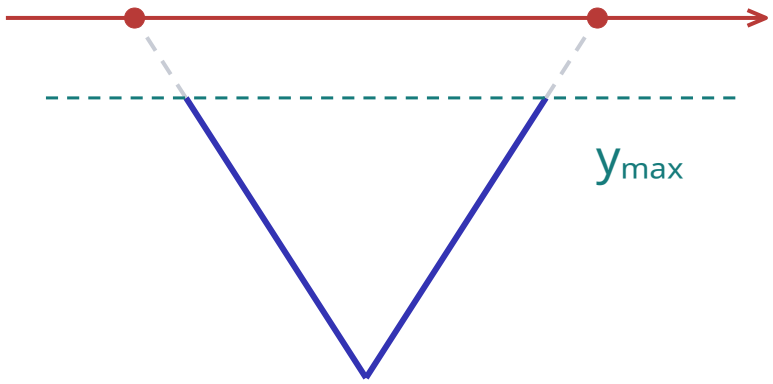
$$P = S + t_r D$$

$$P.y = S.y + t_r D.y$$

Keep $t_r \geq 0$ and $0 \leq P.y < h - r^2/h$.

Apply the height test to the hit point P.

Reject both roots:
P.y exceeds y_{max}.



Q2(d) Cone-intersection algorithm

```
def coneIntersect(S, D, α, r, h, t):
    k = tan(α)**2
    A = D.x**2 + D.z**2 - k*D.y**2
    B = 2*(S.x*D.x + S.z*D.z - k*S.y*D.y)
    C = S.x**2 + S.z**2 - k*S.y**2
    hits = []
    for tr in distinctRealRoots(A, B, C):
        P = S + tr*D
        if tr >= 0 and 0 <= P.y < h - r*r/h:
            hits.append(tr)
    t.extend(hits)
    t.sort()
    return len(hits)
```

distinctRealRoots checks Δ and handles the linear case A = 0.
It returns a repeated tangent root once.

Q2(d) Grading

4 marks, plus the stated quadratic bonus

1	Correctly substitutes $S + t_r D$ into the cone equation.
2	Filters intersections using the correct cone height.
1	Returns the number of valid intersections.

List interpretation

Additional +0.5 for populating t and +0.5 for sorting it.

These list credits cannot exceed the question's full marks.

Separate bonus: +1 for checking the discriminant
or deriving the correct exact intersection parameters.

Q2(d) Partial credit and common mistakes

2 / 2	Correct height check on the hit point $P = S + t_r D$.
1.5 / 2	Uses $P.y < h$.
1 / 2	Height check is wrong but consistent with Q2(b).
−0.5	Uses the correct bound on the wrong quantity, e.g. t_r instead of the hit point's y-coordinate.
Accepted	Height filtering performed in Q2(e).
1	Returned count is supported by earlier intersection logic, such as an equation for calculating the hits.
−0.5	Calculates the count but does not return it.

Q2(e) Combining sphere and cone hits

3 marks

```
def iceCreamIntersect(S, D, α, r, h, t):
    sphereHits = []
    coneHits = []
    ns = sphereIntersect(S, D, r, h, sphereHits)
    nc = coneIntersect(S, D, α, r, h, coneHits)
    t.extend(sphereHits)
    t.extend(coneHits)
    t.sort()
    return ns + nc
```

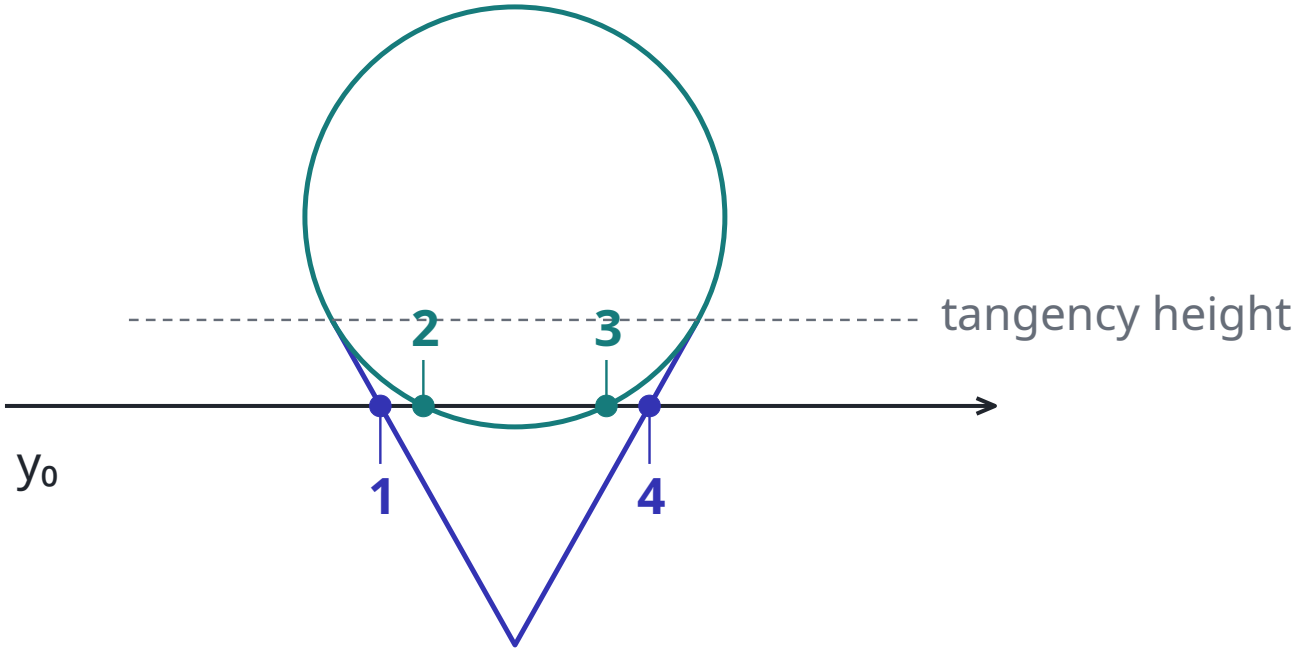
Separate lists keep the calls independent. A shared list also works if both routines append and preserve existing hits.

Q2(f) Four intersections

1 mark

2 cone hits + 2 sphere hits = 4

Choose a horizontal ray below tangency and above the bottom of the sphere: $h - r < y_0 < h - r^2/h$.



Cone: 1, 4 Sphere: 2, 3

Q2(e)–(f) Grading

1	Q2(e): correct sphereIntersect and coneIntersect calls.
2	Q2(e): returns the correct count. Count calculated but not returned: −1.
+0.5 each	Populates t and sorts t. These additional credits cannot exceed the question's full marks.
Shared t	A shared t requires appending in Q2(d). With no implementation given, reinitialization was assumed.
Accepted	Removing duplicate hits at the tangency circle.
1	Q2(f): four intersections and a correct drawing.
0.5	Q2(f): multiple drawings, at least one correct.

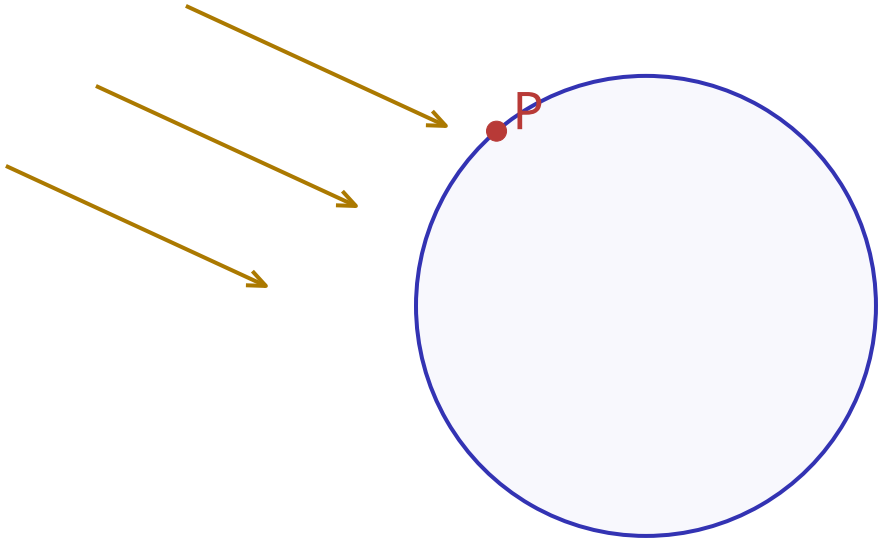
Q3(a) Highlight on an ideal mirror

2 marks

A directional light illuminates an ideally specular sphere.

Fix one illuminated surface point P.

How many eye positions can see P as a highlight?



Q3(a) Infinitely many eye positions

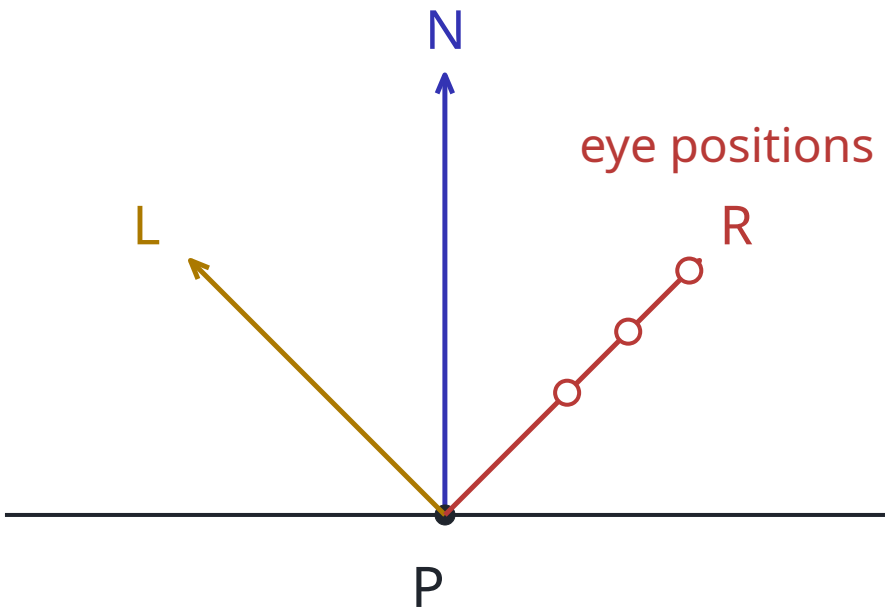
$$R = 2(N \cdot L)N - L$$

An ideal mirror reflects light in exactly one direction R.

Every eye $E = P + sR$, $s > 0$, can see P as a highlight.

$$V = (E - P)/\|E - P\| = R$$
$$V \cdot R = 1$$

L points toward the light.
Each eye looks back toward P.



Q3(a) Grading

2 marks

1	Ideal specular reflection sends light in exactly one direction.
1	Infinitely many eye positions lie along reflected direction R from the fixed point P.
1 total	A correct answer that fixes a special L, e.g. L = V = R.

Lambert's law describes ideal diffuse reflection.

Phong and Blinn approach the ideal specular case as the shininess exponent tends to infinity.

Q3(b) Phong and Blinn dot products

2 marks

Phong: $V \cdot R$

Blinn: $N \cdot H, \quad H = (L + V)/\|L + V\|$

Choose unit vectors with different dot products.

Keep L and V above the surface.
Avoid zero vectors and grazing degeneracies.

A specified angle for each vector removes ambiguity.

Q3(b) A nondegenerate configuration

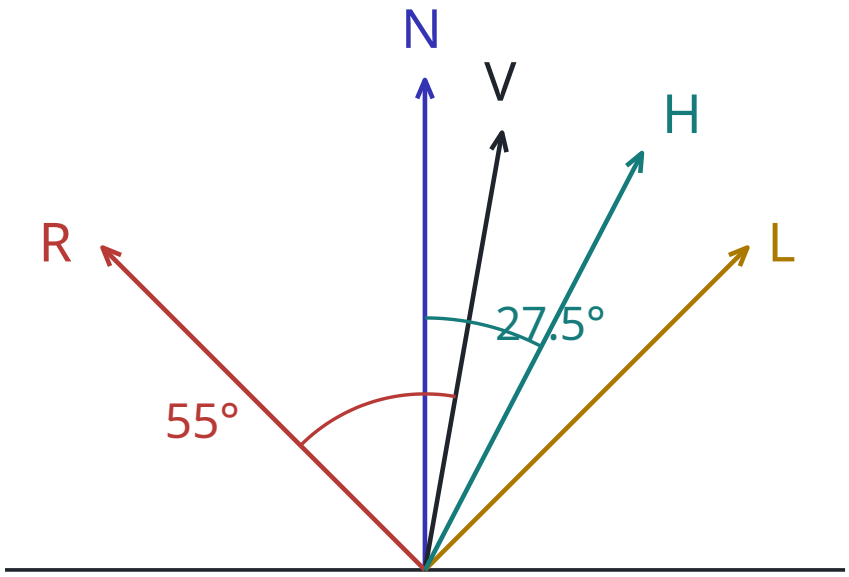
Angles measured from N in the same plane:

L: +45° V: +10° R: −45° H: +27.5°

$V \cdot R = \cos 55° \approx 0.574$

$N \cdot H = \cos 27.5° \approx 0.887$

N = (0, 1, 0)
L = (sin 45°, cos 45°, 0)
V = (sin 10°, cos 10°, 0)



Q3(b) Grading

2 marks

2	A valid configuration, drawn or given numerically.
1	Otherwise valid configuration with degenerate vectors, e.g. zero vectors or directions below the surface.
1	Relative angles give different dot products, but degenerate configurations remain possible.
0	L and V lie on opposite sides of N with unspecified angles. Angles H–N and V–R alone do not resolve this.

V, L, and N must be unit vectors.
Grading calculations used normalized vectors.

Q3(c) Refraction through glass

2 marks

a is the refractive index of air. i is the index of glass.

$$a \sin \theta_a = i \sin \theta_i$$

At entry, the ray bends toward the surface normal.

Does this imply i > a? Explain using Snell’s law.

When is the exiting ray parallel to the entering ray?

Angles are measured from the normal, not from the surface.

Q3(c) Index and parallel exit ray

At oblique entry, $\theta_i < \theta_a$, so

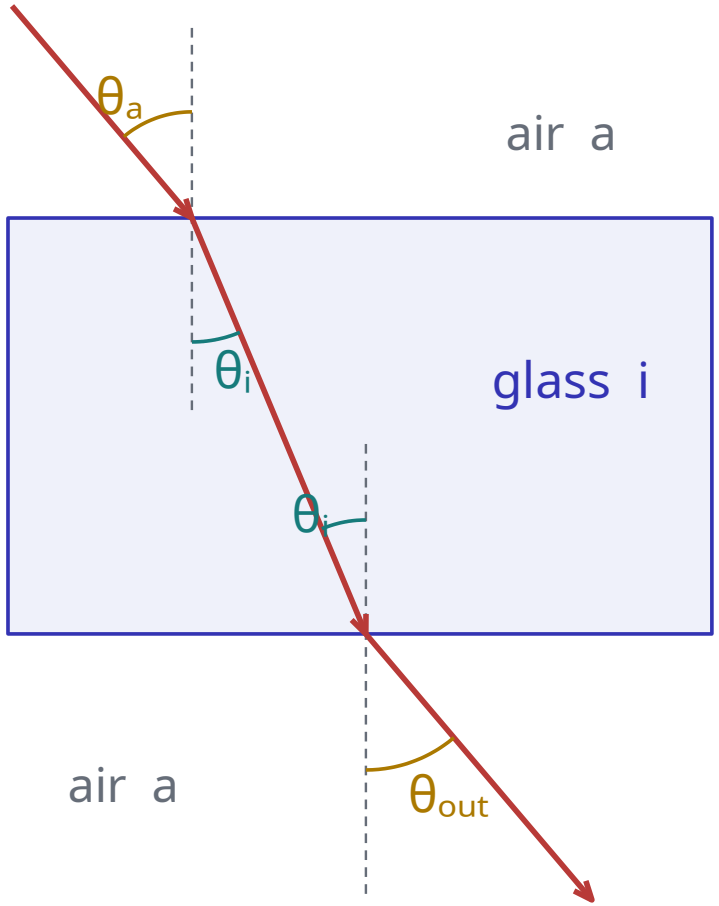
$\sin \theta_i < \sin \theta_a$.

$i / a = \sin \theta_a / \sin \theta_i > 1$

For parallel entry and exit faces, the internal angle at exit is also θ_i.

$i \sin \theta_i = a \sin \theta_{out}$

$\theta_{out} = \theta_a$



Q3(c) Grading

2 marks

1	States i > a and connects the entry-angle inequality to Snell's law.
1	Explicitly states that the glass's entry and exit surfaces are parallel.
Accepted	i = a as a condition for parallel entry and exit rays.
Insufficient	Snell's law plus "bends toward the normal" alone does not explain the comparison of refractive indices.
Insufficient	Specifying only the exit medium's index. The exit surface normal also matters.

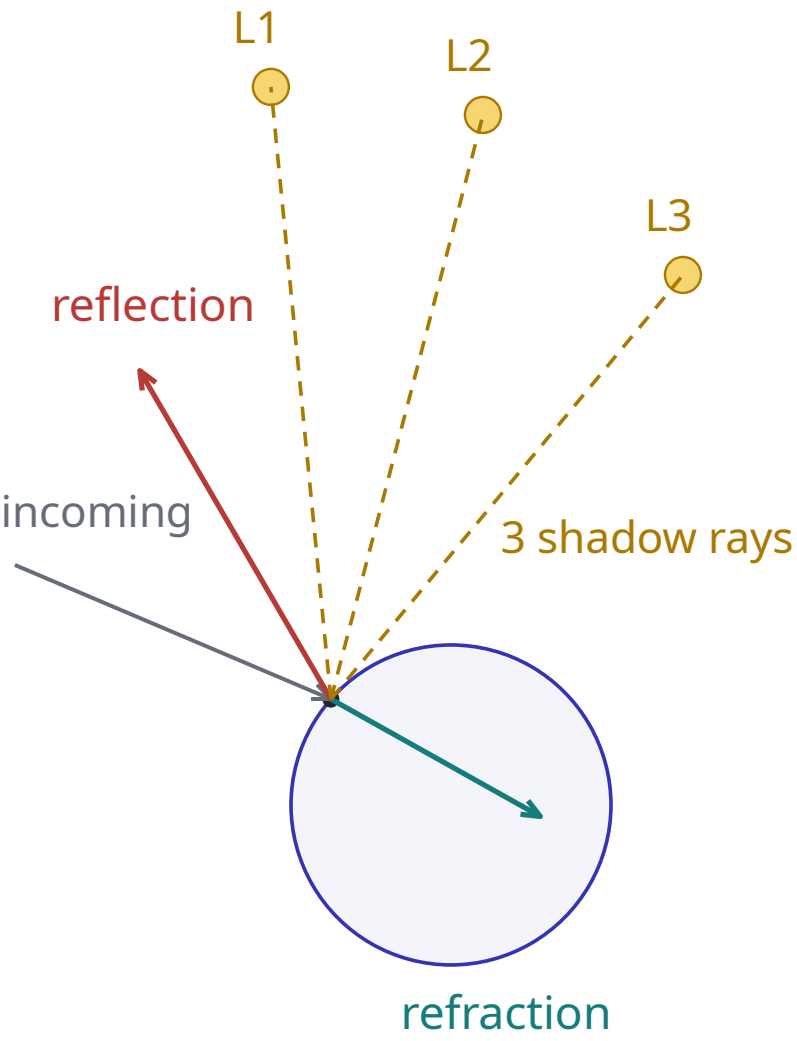
Q3(d) Five new rays

2 marks

1 reflection ray

1 refraction ray

3 shadow rays, one to each light



1 + 1 + 3 = 5

The incoming eye ray is already present.
Count the new rays spawned at the hit.

Q3(d) Grading

2 marks

1	Counts 2 rays: reflection and refraction.
1	Explicitly counts 3 shadow rays.
Required	Each count has a clear cause, e.g. “3 shadow rays, one per light”.
0.5	States the correct components, but gives the wrong final total, such as 6.

2 + 3 = 5 new rays